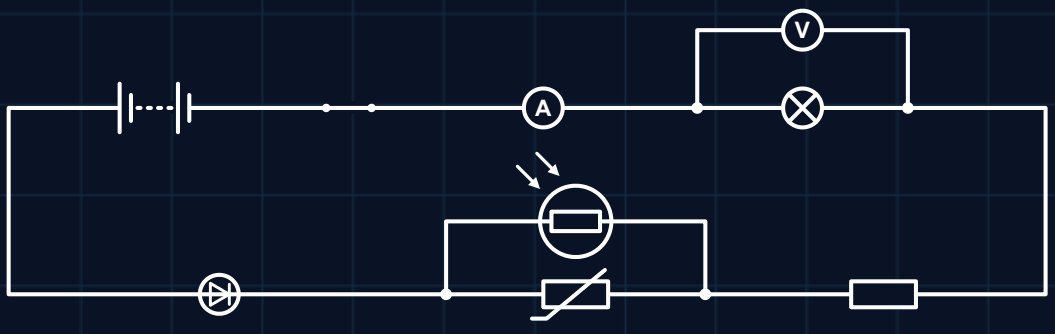
YEAR 10 PHYSICS, AQA GCSE, UNIT REVIEW

# ELECTRIC CIRCUITS

## UNIT REVIEW AFTER LESSON 12

Mixed exam practice from the whole unit, a fix-it log, and a static electricity page for GCSE Physics students.




NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

REFERENCE

# EQUATION TOOLKIT

You will meet these one at a time. Learn each equation **in words** as well as symbols: AQA gives you an equation sheet in the exam (it does not include $R_{total} = R_1 + R_2$), but knowing the equations by heart makes every question faster.

## $Q = I\,t$

Lesson 2

**charge flow = current × time**

C, A, s

Rearranged: I = Q ÷ t and t = Q ÷ I

## $E = Q\,V$

Lesson 3

**energy transferred = charge × p.d.**

J, C, V

Rearranged: V = E ÷ Q and Q = E ÷ V

## $V = I\,R$

Lesson 4

**p.d. = current × resistance**

V, A, Ω

Rearranged: I = V ÷ R and R = V ÷ I

## $R_{total} = R_1 + R_2$

Lesson 6

**total resistance in series**

Ω

Rearranged: (series only)

## $P = V\,I$

Lesson 11

**power = p.d. × current**

W, V, A

Rearranged: V = P ÷ I and I = P ÷ V

## $P = I^2\,R$

Lesson 11

**power = current² × resistance**

W, A, Ω

Rearranged: R = P ÷ I² and I = √(P ÷ R)

## $E = P\,t$

Lesson 11

**energy transferred = power × time**

J, W, s

Rearranged: P = E ÷ t and t = E ÷ P

## $V_p\,I_p = V_s\,I_s$

Lesson 12

**transformer: power in = power out**

V, A

Rearranged: $I_s = V_p\,I_p \div V_s$

## FOUR STEPS FOR EVERY CALCULATION

**1**
Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

**2**
Write the equation you will use, in symbols.

**3**
Substitute the numbers, then rearrange if needed.

**4**
Give the answer with a unit. Ask: is it sensible?

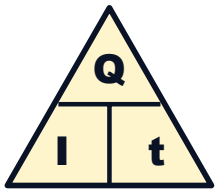
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| CONVERT | HOW | EXAMPLE |
|---|---|---|
| minutes → seconds | × 60 | 3.0 min = 180 s |
| hours → seconds | × 3600 | 2.0 h = 7200 s |
| milliamps (mA) → amps (A) | ÷ 1000 | 250 mA = 0.25 A |
| kilowatts (kW) → watts (W); kilohms (kΩ) → ohms (Ω) | × 1000 | 2.2 kW = 2200 W |
| kilovolts (kV) → volts (V) | × 1000 | 400 kV = 400 000 V |

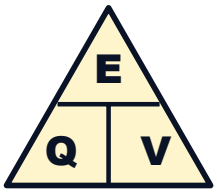
## EQUATION TRIANGLES: HELP WITH REARRANGING

1. Cover the quantity you want to find.
2. The two left **side by side**? **Multiply** them.
3. One **above** the other? **Divide** the top by the bottom.

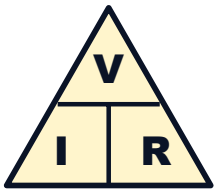
Example: to find I from Q = I t, cover I. Q is above t, so I = Q ÷ t. Triangles only help with rearranging: you still need to learn each equation. For P = I² R, the triangle gives I²; take the square root to find I.



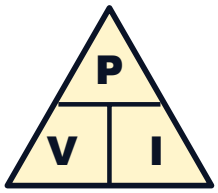

**Q = I t**
Q = I × t
I = Q ÷ t
t = Q ÷ I



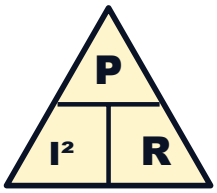

**E = Q V**
E = Q × V
Q = E ÷ V
V = E ÷ Q



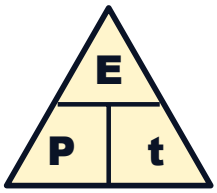

**V = I R**
V = I × R
I = V ÷ R
R = V ÷ I




**P = V I**
P = V × I
V = P ÷ I
I = P ÷ V




**P = I² R**
P = I² × R
R = P ÷ I²
I² = P ÷ R, so I = √(P ÷ R)




**E = P t**
E = P × t
P = E ÷ t
t = E ÷ P

AFTER LESSON 12, AT HOME OR IN CLASS

# UNIT REVIEW

Mixed questions from the whole unit, like a real exam. Try each question without looking back. Then mark it with your teacher's answers and fill in the “fix it” log.

RETRIEVE

## EQUATION RECALL CHALLENGE

Cover the toolkit page first.

**1** Write each equation in symbols. Then give the unit of every quantity in it.

| Equation in words | In symbols | Units |
|---|---|---|
| **charge flow = current × time** | | |
| **energy transferred = charge × p.d.** | | |
| **p.d. = current × resistance** | | |
| **total resistance in series** | | |
| **power = p.d. × current** | | |
| **power = current² × resistance** | | |
| **energy transferred = power × time** | | |

AQA-STYLE

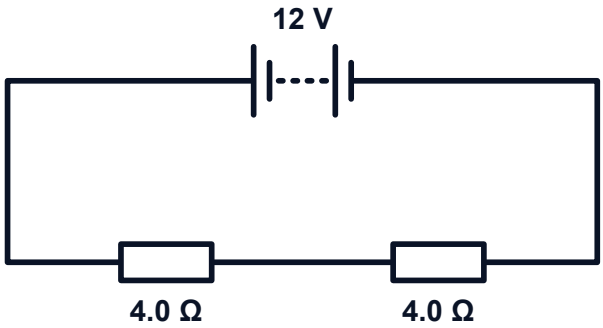
## MIXED EXAM PRACTICE

41 marks, about 50 minutes

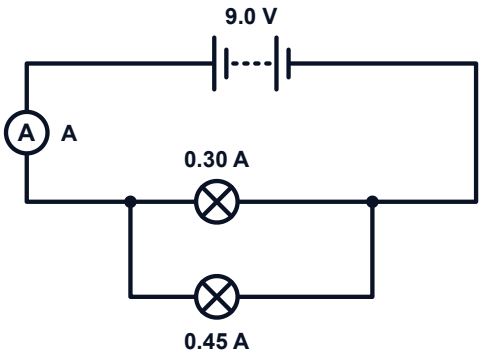
**2** A current of 0.80 A flows through a lamp for 2.5 minutes. Calculate the charge that flows. **[3 marks]**

**3** A component transfers 48 J of energy when 8.0 C of charge passes through it. (a) Calculate the p.d. across it. (b) State what this p.d. means. **[3 marks]**

**4** Calculate the total resistance, the current, and the p.d. across one resistor. **[4 marks]**




**5** The lamps are in parallel. State the p.d. across each lamp and the reading on ammeter A. Give reasons. **[3 marks]**




**6** In the wire-length practical, one point on the graph of resistance against length is well above the line. (a) Suggest two possible causes. (b) What should the student do? **[4 marks]**

**7** Sketch the I–V graph of a diode. Explain how it differs from the graph of a fixed resistor. **[4 marks]**

**8** A thermistor is connected in series with an ammeter and a battery. Explain how the ammeter reading changes as the thermistor cools down. **[3 marks]**

**9** A kettle has a metal case connected to the earth wire. Explain why the earth wire is needed, including the role of the fuse. **[5 marks]**

**10** A 1.5 kW heater runs on the 230 V mains for 4.0 minutes. Calculate (a) the energy transferred and (b) the current. **[5 marks]**

**11** Use the equations $P = V I$ and $P = I^2 R$ to explain why the National Grid transmits electricity at a very high p.d. **[4 marks]**

**12** An ideal step-up transformer has 25 kV and 1200 A in the primary coil. The secondary p.d. is 400 kV. Calculate the secondary current. **[3 marks]**

REVIEW LATER

# FIX-IT LOG

Turn every lost mark into a target.

| Q | What went wrong? | The correct physics (or method) is… | Lesson to revisit |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

TRIPLE SCIENCE ONLY (AQA 4.2.5)

# EXTENSION: STATIC ELECTRICITY

Only for students taking GCSE Physics (separate science). Combined Science students can skip this page.

When two insulating materials are rubbed together, **electrons** are transferred from one to the other. The one that **gains electrons** becomes **negatively** charged; the one that **loses electrons** is left **positively** charged. Only electrons move, never positive charge.

**Like charges repel; unlike charges attract.** These are non-contact forces: they act without the objects touching.

A charged object creates an **electric field** around it. Another charged object in the field feels a force. The field is **strongest close to the object** and gets weaker further away.

If enough charge builds up, the p.d. between the object and the earthed conductor nearby becomes large enough for charge to jump across the air: a **spark**.

**1** A plastic rod becomes negatively charged when it is rubbed with a cloth. Which way did the electrons move? What charge is left on the cloth? **[2 marks]**

**2** Draw the electric field around an isolated, positively charged sphere. Add arrows, and show where the field is strongest. **[3 marks]**

**3** A small negatively charged sphere is moved towards a large positively charged sphere. Explain what happens to the force on the small sphere, without the spheres touching. **[4 marks]**

REFERENCE

# KEY WORDS

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

**Alternating current (ac)**: Current that keeps changing direction. UK mains is ac at 50 Hz.

**Ammeter**: Measures current. Connected in series.

**Ampere (A)**: Unit of current. 1 A = 1 C per second.

**Anomaly**: A result that does not fit the pattern of the others.

**Battery**: Two or more cells joined together.

**Charge (Q)**: Carried by particles such as electrons. Unit: coulomb (C).

**Circuit breaker**: A safety switch that cuts off the current if it gets too large.

**Conventional current**: The agreed direction of current: from + to −.

**Current (I)**: The rate of flow of charge.

**Diode**: Lets current flow in one direction only.

**Direct current (dc)**: Current that flows in one direction only, e.g. from a cell.

**Directly proportional**: If one quantity doubles, the other doubles. The graph is a straight line through the origin.

**Earth wire**: Green and yellow safety wire; carries a current only if there is a fault.

**Electron**: A tiny negatively charged particle. Free electrons carry the current in metals.

**Filament lamp**: A lamp with a thin metal filament; its resistance increases as it heats up.

**Fuse**: A thin wire that melts and breaks the circuit if the current is too large.

**I–V characteristic**: A graph of current against p.d. for a component.

**LDR**: Light-dependent resistor: resistance decreases as light intensity increases.

**LED**: Light-emitting diode: a diode that gives out light.

**Live wire**: Brown wire at about 230 V; dangerous to touch.

**National Grid**: The system of cables and transformers linking power stations to consumers.

**Neutral wire**: Blue wire that completes the circuit; at or close to 0 V.

**Ohm (Ω)**: Unit of resistance.

**Ohmic conductor**: Current is directly proportional to p.d. at constant temperature.

**Parallel**: Components on separate branches. A voltmeter is connected in parallel.

**Potential difference (V)**: Energy transferred per unit charge between two points. Unit: volt (V).

**Power (P)**: Energy transferred per second. Unit: watt (W).

**Resistance (R)**: How much a component opposes the current. Unit: ohm (Ω).

**Series**: Components connected one after another in a single loop.

**Thermistor**: Resistance decreases as temperature increases.

**Transformer**: Changes the size of an alternating p.d. (step-up or step-down).

**Volt (V)**: Unit of p.d. 1 V = 1 J per C.

**Voltmeter**: Measures p.d. Connected in parallel, across a component.

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FINISH

# PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

| I CAN… | R / A / G |
|---|---|
| **01: Complete circuits and symbols** | |
| explain why a circuit must be a complete loop | |
| recognise and draw the symbols for a cell, battery, switch, lamp, resistor and ammeter | |
| draw a neat circuit diagram using the rules | |
| connect an ammeter in series | |
| **02: Current and charge** | |
| describe current as the rate of flow of charge | |
| calculate charge, current or time using Q = I t | |
| convert minutes to seconds before calculating | |
| explain why current is not used up in a loop | |
| **03: Potential difference** | |
| explain p.d. as energy transferred per coulomb | |
| connect a voltmeter in parallel | |
| use E = Q V and its rearrangements | |
| use ‘p.d.s add up’ in a series loop | |
| **04: Resistance and V = I R** | |
| describe resistance and state its unit | |
| use V = I R and its rearrangements | |
| convert mA and kΩ | |
| decide from data or a graph whether a component is ohmic | |
| explain why heating increases resistance | |
| **05: Resistance of a wire** | |
| draw and set up the circuit to find a resistance | |
| name the independent, dependent and control variables | |
| plot a graph and draw a line of best fit | |

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| I CAN… | R / A / G |
|---|---|
| spot an anomaly and suggest a cause | |
| **06: Series and parallel circuits** | |
| tell series and parallel circuits apart | |
| use the current and p.d. rules in series and parallel | |
| calculate total resistance, current and p.d.s in a series circuit | |
| explain why parallel branches reduce the total resistance | |
| **07: Resistors in series and parallel** | |
| set up series and parallel combinations with the meters in the right places | |
| calculate total resistance from V and I | |
| explain why series adds and parallel lowers resistance | |
| suggest a specific improvement to a practical | |
| **08: I–V graphs** | |
| read values from an I–V graph, including negative values | |
| sketch the I–V graphs for a fixed resistor, lamp and diode | |
| calculate resistance at a point | |
| explain why the lamp graph curves | |
| **09: Investigating I–V characteristics** | |
| set up the I–V circuit and explain each part | |
| take and record readings in both directions | |
| plot an I–V graph with the origin in the middle | |
| describe the extra steps for a diode | |
| **10: Thermistors, LDRs and sensors** | |
| describe how a thermistor responds to temperature | |
| describe how an LDR responds to light | |
| explain how the current in a sensor circuit changes | |
| choose and justify a sensor for a job | |
| **11: Power and energy transfer** | |
| describe power as energy transferred per second | |

| I CAN… | R / A / G |
|---|---|
| use E = P t with unit conversions | |
| use P = V I and P = I² R | |
| choose the right equation from the data given | |
| **12: Mains, safety and the National Grid** | |
| describe the difference between ac and dc | |
| state the UK mains p.d. and frequency | |
| name the three wires and their colours and jobs | |
| explain how the earth wire and fuse protect a user | |
| explain why the grid uses step-up and step-down transformers | |